

# COORDINATE-BLOCK FACTORIZATION UNDER SINGLE-SANDWICH DIVISION AND APPLICATIONS TO PROPERTY $(TT)/T$

GPT-5.6 SOL

ABSTRACT. We prove a coordinate-block factorization theorem for general linear groups over rings satisfying the single-sandwich condition

$$a \neq 0 \implies xay = 1 \text{ for some } x, y.$$

Every element of  $GL_n(R)$  is a product of at most  $2n + 4$  factors, each an elementary transvection or an element of one fixed coordinate copy of  $GL_{n-1}(R)$ . Thus  $GL_n(R)$  has width at most  $2n + 4$  relative to the elementary root subgroups and that fixed coordinate block.

The algebraic factorization is the main input developed here. We combine it with Mimura's relative quasi-cocycle rigidity framework for elementary groups over finite free associative  $\mathbb{F}_2$ -algebras. If  $R$  is a nontrivial finite-type  $\mathbb{F}_2$ -algebra carrying a binary Leavitt family, satisfying the single-sandwich condition, and satisfying  $\text{diag}(u, 1) \in E_2(R)$  for every  $u \in R^\times$ , then  $E_n(R)$  has property  $(TT)/T$  for every  $n \geq 2$ . Consequently,

$$E_n(L_{\mathbb{F}_2}(1, 2)) \text{ has property } (TT)/T \quad (n \geq 2).$$

We also give a quantitative reconstruction of the relative root estimate. Together with OpenAI's construction of the first nonsofic group, whose example is the binary Leavitt unit group [11, Theorem 1.1, Chapter 3], the result gives explicit nonsofic groups with property  $(TT)/T$ .

## 1. INTRODUCTION

Property  $(TT)/T$ , introduced by Mimura, asks that Hilbert-space quasi-cocycles be bounded after the invariant-vector part of the representation has been removed. It is stronger than Kazhdan's property  $(T)$  and is designed to control bounded cohomology and homomorphisms into groups admitting unbounded quasi-cocycles [8]. Ershov and Jaikin-Zapirain proved property  $(T)$  for elementary groups over arbitrary finitely generated associative rings [4]. The corresponding quasi-cocycle problem is subtler: the bounded error in the cocycle identity accumulates under a long word, so finite generation alone gives no global bound.

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The present paper supplies a globalization mechanism for a class of purely infinite rings. Its algebraic input is an explicit matrix factorization. The hypothesis is the following strong two-sided division property:

$$(1) \quad a \neq 0 \implies \exists x, y \in R, \quad xay = 1.$$

We call (1) the single-sandwich property. Ara, Goodearl, and Pardo proved that, for a nondivision simple ring, this is equivalent to pure infiniteness [3, Theorem 1.6].

**Theorem A** (Coordinate-block factorization). *Let  $R$  be a nontrivial ring satisfying (1), let  $n \geq 2$ , and fix a coordinate  $j < n$ . Every  $A \in \mathrm{GL}_n(R)$  admits a factorization of length at most  $2n + 4$  in which every factor is either an elementary transvection or belongs to the coordinate copy*

$$\mathrm{GL}_{n-1}^{(j)}(R) = \{B \in \mathrm{GL}_n(R) : Be_j = e_j, e_j^t B = e_j^t\}.$$

Theorem A is a relative bounded-width statement rather than bounded elementary generation: the bounded factors come from the elementary root subgroups together with one fixed coordinate block. This is exactly the form required by the quasi-cocycle argument. By the characterization just cited, the conclusion applies in particular to every unital purely infinite simple ring.

For the synthesis theorem, a *binary Leavitt family* in  $R$  means elements  $x_0, x_1, y_0, y_1$  satisfying

$$y_i x_j = \delta_{ij}, \quad x_0 y_0 + x_1 y_1 = 1.$$

It supplies the matrix self-similarity used to move between ranks. We say that an  $\mathbb{F}_2$ -algebra is *finite type* when it is generated by a finite set as an  $\mathbb{F}_2$ -algebra, equivalently when it is a quotient of  $\mathbb{F}_2\langle X \rangle$  for some finite set  $X$ .

**Theorem B** (Rigidity synthesis). *Let  $R$  be a nontrivial finite-type  $\mathbb{F}_2$ -algebra. Assume that*

- (1)  *$R$  contains a binary Leavitt family;*
- (2)  *$R$  satisfies the single-sandwich property (1);*
- (3)  *$\mathrm{diag}(u, 1) \in E_2(R)$  for every  $u \in R^\times$ .*

*Then  $E_n(R)$  has property  $(TT)/T$  for every  $n \geq 2$ .*

The assumptions play separate roles: finite type presents the ring as a quotient of a finite free algebra, the Leavitt family moves between matrix ranks, the single-sandwich property supplies bounded matrix reduction, and the diagonal condition identifies elementary and general linear groups.

**Theorem C** (Binary Leavitt algebra). *For every  $n \geq 2$ ,*

$$E_n(L_{\mathbb{F}_2}(1, 2)) \quad \text{has property } (TT)/T.$$

OpenAI proved that  $L_{\mathbb{F}_2}(1, 2)^\times$  is nonsofic [11, Theorem 1.1 of Chapter 3]. Combining that theorem with Theorem C gives the following consequence. The same nonsoficity mechanism has since been developed by Kun–Thom and Fournier-Facio [7, 6].

**Corollary 1.1.** *For every  $n \geq 2$ , the group  $E_n(L_{\mathbb{F}_2}(1, 2))$  is nonsofic and has property  $(TT)/T$ .*

*Proof.* Theorem C supplies property  $(TT)/T$ . For nonsoficity, OpenAI's theorem gives that  $L^\times$  is nonsofic [11, Theorem 1.1 of Chapter 3]. The map

$$L^\times \longrightarrow \mathrm{GL}_n(L), \quad u \longmapsto \mathrm{diag}(u, 1, \dots, 1),$$

is injective. Proposition 5.1, followed by the standard block inclusion from rank two to rank  $n$ , shows that its image lies in  $E_n(L)$ . Since every subgroup of a sofic group is sofic,  $E_n(L)$  cannot be sofic.  $\square$

**Relation to previous work.** The argument combines several established theories. Mimura introduced property  $(TT)/T$ , developed the relative-to-global strategy for quasi-cocycles, and proved the relative results for universal-lattice settings used below [8, 9]. Ershov and Jaikin-Zapirain proved property  $(T)$  for elementary groups over finitely generated associative rings [4]. On the ring-theoretic side, Ara, Goodearl, and Pardo established the GE and unit-group  $K_1$  structure of purely infinite simple rings [3], while Ara, Brustenga, and Cortiñas computed the stable algebraic  $K$ -theory of Leavitt path algebras [2].

The pivot in Section 3 belongs to the same Gaussian-elimination circle as the GE reduction of Ara–Goodearl–Pardo. The fixed-coordinate organization is the additional feature used here: the coordinate is chosen before the matrix, exactly one block factor occurs, and the remaining word has the uniform length  $2n + 3$ . Thus the full factorization has width  $2n + 4$ . The rigidity theorem uses this factorization to globalize relative quasi-cocycle bounds. Section 2 gives a quantitative proof of the root estimate, and Section 5 verifies the remaining structural hypotheses for the binary Leavitt algebra.

**Organization.** Section 2 develops the relative root estimate. Section 3 proves the five-move pivot and the coordinate-block factorization. Section 4 combines the analytic and algebraic inputs to prove Theorem B, and Section 5 verifies its hypotheses for the binary Leavitt algebra. The final section discusses the natural extensions suggested by the proof.

## 2. QUASI-COCYCLES AND RELATIVE CONTROL

Let  $G$  be a group and let  $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$  be a unitary representation on a complex Hilbert space. A map  $b : G \rightarrow \mathcal{H}$  is a *quasi-cocycle of defect at most  $D$*  if

$$(2) \quad \|b(gh) - b(g) - \pi(g)b(h)\| \leq D \quad (g, h \in G).$$

An exact cocycle has  $D = 0$ ; it is the translation part of an affine isometric action. A quasi-cocycle permits uniformly bounded failure of the affine action law.

**Definition 2.1.** A group  $G$  has property  $(TT)/T$  if, for every unitary representation  $\pi$  with no nonzero invariant vector, every Hilbert-space quasi-cocycle into  $\pi$  is bounded on  $G$ . For a subgroup  $H \leq G$ , the pair  $(G, H)$  has relative property  $(TT)/T$  if the same hypotheses imply boundedness on  $H$ . We write relative  $(TT)$  when the conclusion holds without excluding invariant vectors.

Mimura's relative input. Let  $X$  be finite and  $A = \mathbb{F}_2\langle X \rangle$ . The pair  $(E_4(A), X_{03})$  has relative property  $(TT)/T$ . Indeed, Mimura proves relative  $(TT)/T$  for the noncommutative universal lattice pair in [9, p. 154, Proposition 9.2.8]:

$$E_m(\mathbb{Z}\langle X \rangle) \geq E_{m-1}(\mathbb{Z}\langle X \rangle), \quad m \geq 3.$$

Relative  $(TT)/T$  descends through a surjective homomorphism: pull a representation and quasi-cocycle back along the quotient map and then apply the relative bound. Reduce coefficients modulo 2, take  $m = 4$ , and conjugate the standard rank-three block so that it contains  $X_{03}$ . Boundedness on that block implies boundedness on the root subgroup.

The local input used below is slightly stronger than a relative  $(TT)/T$  statement because no hypothesis is imposed on invariant vectors.

We first isolate the quantitative rank-three estimate used in the direct proof. For distinct  $i, j, k$ , write  $H_{ijk} = \langle X_{ik}, X_{jk} \rangle \leq E_3(A)$ , and identify the elements of  $X$  with the corresponding free generators of  $A$ .

The proof uses the word-length filtration  $A_{\leq d} \subseteq A$ . Put  $V_d = A_{\leq d} \oplus A_{\leq d}$ , viewed as the finite elementary abelian subgroup

$$(a, c) \longmapsto x_{ik}(a)x_{jk}(c) \quad \text{of } H_{ijk}.$$

For a unitary representation  $\sigma$  and  $z \in \mathcal{H}$ , let  $z_{d,\chi}$  be the Fourier component of  $z$  belonging to  $\chi \in \widehat{V}_d$ . The components are orthogonal and their squared norms sum to  $\|z\|^2$ . Write a character as  $(\chi_1, \chi_2)$ , and let  $\nu(\chi_r)$  be the least word degree on which  $\chi_r$  is nontrivial, with value  $d+1$  for the trivial character. The nontrivial pairs split into

$$\begin{aligned} A & 0 < \nu(\chi_2) < \nu(\chi_1) \\ B & 0 < \nu(\chi_1) = \nu(\chi_2) \\ C & 0 < \nu(\chi_1) < \nu(\chi_2) \\ D & \nu(\chi_1) = 0 \text{ or } \nu(\chi_2) = 0. \end{aligned}$$

Let  $m_d^A, m_d^B, m_d^C, m_d^D$  be the corresponding Fourier masses. Let  $\partial_{1,d}$  and  $\partial_{2,d}$  be the masses for which the indicated coordinate character is trivial on  $A_{\leq d-1}$  and becomes nontrivial on  $A_{\leq d}$ .

**Lemma 2.2** (Finite-plane energy estimate). *If  $P$  is the orthogonal projection onto  $\mathcal{H}^{H_{ijk}}$  and  $w = z - Pz$ , then*

$$\begin{aligned} \|w\|^2 & \leq d_{ik,1}^2 + d_{jk,1}^2 + \frac{3}{2} \sum_{s \in X} 2 \|z\| (d_{ij,s} + d_{ji,s}) \\ (3) \quad & + 2 \|z\| (d_{ij,1} + d_{ji,1}), \end{aligned}$$

where  $d_{ab,t} = \|\sigma(x_{ab}(t))z - z\|$ .

*Proof.* We apply the Fourier argument to the underlying real Hilbert space. The unitary operators are orthogonal there, and all norms, fixed spaces, and displacements are unchanged. Set

$$U_{ab} = 2\|z\| d_{ab,1}, \quad G_{ab} = \sum_{s \in X} 2\|z\| d_{ab,s}.$$

Conjugation by  $x_{ji}(1)$  shears region  $A$  into region  $B$ , and conjugation by  $x_{ij}(1)$  does the same from  $C$  to  $B$ . Fourier orthogonality and

$$\left| \sum_{\chi \in \Omega} \|z_{d,\chi}\|^2 - \sum_{\chi \in \Omega} \|z'_{d,\chi}\|^2 \right| \leq (\|z\| + \|z'\|) \|z - z'\|$$

for  $z' = \sigma(g)z$  give

$$(4) \quad m_d^A \leq m_d^B + U_{ji}, \quad m_d^C \leq m_d^B + U_{ij}.$$

On region  $D$ , one coordinate character detects the unit coefficient. The nontrivial character of  $\mathbb{F}_2$  has character gap 4, so the negative-character projection formula for an involution gives

$$(5) \quad m_d^D \leq \frac{1}{4}(d_{ik,1}^2 + d_{jk,1}^2).$$

We next record the cross-stage inequalities controlling the equal-valuation region. If  $m_d^S$  denotes the mass in a union  $S$  of valuation regions, restriction of characters from  $V_{d+1}$  to  $V_d$  gives

$$(6) \quad m_d^S \leq m_{d+1}^S + \partial_{1,d+1} + \partial_{2,d+1}.$$

The two generator shears give the sharper descents

$$(7) \quad m_{d+2}^{A \cup B} \leq m_{d+1}^{C \cup D} + G_{ji} + \partial_{2,d+2},$$

$$(8) \quad m_{d+2}^{C \cup B} \leq m_{d+1}^{A \cup D} + G_{ij} + \partial_{1,d+2}.$$

We prove the first; the second follows after interchanging the two coordinates. Remove the characters whose second coordinate is new in top degree. Their mass is at most  $\partial_{2,d+2}$ . For each remaining character, choose the least free generator in the leading word of its second coordinate. The opposite shear  $x_{ji}(s)$  lowers that leading degree, and restriction to  $V_{d+1}$  places the resulting character in  $C \cup D$ . The chosen generator is retained in the receiving functional, so the receiving sets for distinct generators are disjoint. Fourier covariance and the variation inequality above charge the source mass to  $m_{d+1}^{C \cup D}$ , with error  $2\|z\| d_{ji,s}$  on the part labeled by  $s$ . Summing over  $s \in X$  proves (7). The forward shear proves (8). Thus generator multiplication is handled by passing between consecutive stages, with the newly exposed terms recorded by the two top-boundary masses.

Add (7) and (8), use (6) to move their coarse masses to stage  $d+2$ , and then use (4) and (5). Collecting the four region coefficients gives

$$m_{d+2}^A + m_{d+2}^B + m_{d+2}^C + m_{d+2}^D \leq d_{ik,1}^2 + d_{jk,1}^2 + \frac{3}{2}(G_{ji} + G_{ij}) + U_{ji} + U_{ij}$$

$$(9) \quad + \frac{9}{2}(\partial_{1,d+2} + \partial_{2,d+2}).$$

This is the finite-stage estimate whose limit is (3); it also accounts for the coefficient 3/2 there.

The boundary limit follows from an exact telescoping identity. If  $T_{r,d}$  is the mass on which the  $r$ -th coordinate character is trivial on  $A_{\leq d}$ , then

$$T_{r,d} = \partial_{r,d+1} + T_{r,d+1}.$$

The nonnegative sequence  $T_{r,d}$  decreases, so  $\partial_{r,d+1} = T_{r,d} - T_{r,d+1} \rightarrow 0$ . Finally,  $V_d$  exhausts  $H_{ijk}$ , and the left side of (9) converges to  $\|z - Pz\|^2$ . Passing to the limit proves (3).  $\square$

**Lemma 2.3** (Localized plane estimate). *Let  $\sigma : E_3(A) \rightarrow \mathcal{U}(\mathcal{H})$  be a unitary representation and let  $z \in \mathcal{H}$ . Suppose  $B \geq 0$  and the displacement of  $z$  is at most  $B$  under*

$$x_{ik}(1), x_{jk}(1), x_{ij}(1), x_{ji}(1), x_{ij}(s), x_{ji}(s) \quad (s \in X).$$

*Then, for every  $g \in H_{ijk}$ ,*

$$\|\sigma(g)z - z\| \leq 2(6|X| + 6)(B + 1).$$

*Proof.* Put  $\delta = B + 1$ , let  $P$  be the orthogonal projection onto  $\mathcal{H}^{H_{ijk}}$ , and set  $w = z - Pz$ . The adjacent root groups  $X_{ij}$  and  $X_{ji}$  normalize  $H_{ijk}$ , while  $X_{ik}, X_{jk} \leq H_{ijk}$ . Orthogonal projection therefore passes each of the displayed displacement bounds from  $z$  to  $w$ , with the strict bound  $\delta$ . Apply Lemma 2.2 to  $w$ . Since  $Pw = 0$ , it gives

$$\|w\|^2 \leq 2\delta^2 + (6|X| + 4)\|w\|\delta.$$

If  $\|w\| \leq \delta$ , the desired bound on  $w$  is immediate. Otherwise  $\delta \leq \|w\|$ , and the last inequality gives  $\|w\| \leq (6|X| + 6)\delta$ . Finally, every  $g \in H_{ijk}$  fixes  $Pz$ , so  $\|\sigma(g)z - z\| \leq 2\|w\|$ .  $\square$

**Lemma 2.4** (Row-plane symmetry). *Let  $^- : A \rightarrow A$  reverse every word and fix the free generators. The inverse-transpose map*

$$\Theta(g) = \bar{g}^{-t}$$

*is an automorphism of  $E_3(A)$ , and*

$$\Theta(x_{ij}(a)) = x_{ji}(-\bar{a}) \quad (a \in A).$$

*In particular,  $\Theta(x_{ij}(t)) = x_{ji}(t)$  for  $t \in \{1\} \cup X$ . Consequently Lemma 2.3 applies with the two root directions reversed.*

*Proof.* Word reversal followed by transpose identifies the matrix ring with its opposite ring. On units, inversion identifies the opposite group with the original group, so  $\Theta$  is an automorphism. Since  $E_{ij}^2 = 0$ ,  $x_{ij}(a)^{-1} = x_{ij}(-a)$ . In characteristic two the sign vanishes, while word reversal fixes 1 and every  $s \in X$ , which gives the displayed formula.  $\square$

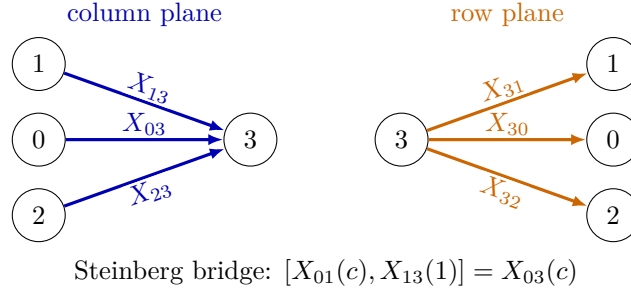


FIGURE 1. The rank-four root geometry. The two rank-three plane estimates control the blue column roots and, through Lemma 2.4, the orange row roots. The displayed Steinberg commutator recovers the target root  $X_{03}$ .

**Theorem 2.5** (Root quasi-cocycle estimate). *Let  $X$  be finite, put  $A = \mathbb{F}_2\langle X \rangle$ , and let*

$$G = E_4(A), \quad X_{03} = \{I + aE_{03} : a \in A\}.$$

*For every unitary representation  $\pi$ , every  $D \geq 0$ , and every quasi-cocycle  $b : G \rightarrow \mathcal{H}$  of defect at most  $D$ , the set  $b(X_{03})$  is bounded.*

*Proof.* Let  $N$  be the last-column subgroup generated by  $X_{03}, X_{13}, X_{23}$ . Decompose

$$\mathcal{H} = \mathcal{H}^N \oplus (\mathcal{H}^N)^\perp, \quad b(r) = p + w$$

for  $r = I + aE_{03}$ . Since the coefficient ring has characteristic two,  $r^2 = 1$ . First,  $\|b(1)\| \leq D$ , by applying (2) to  $1 \cdot 1$ . Apply the same inequality to  $r^2$  and project orthogonally to  $\mathcal{H}^N$ . Since  $r \in N$ , it acts as the identity on this subspace. If  $q$  denotes the projection of  $b(1)$ , contractivity of the projection gives

$$\|q - p - p\| \leq D, \quad \|q\| \leq D.$$

Thus  $\|2p\| \leq 2D$ , and hence

$$(10) \quad \|p\| \leq D.$$

For the moving component, let  $Q_X \subseteq E_3(A)$  consist of all elementary roots with coefficient 1 or one of the free generators. Embed it on the coordinates  $\{1, 2, 3\}$ , and also on the coordinates  $\{0, 1, 2\}$  using the transpose-word-reversal symmetry of Lemma 2.4; the geometry is shown in Figure 1. Put

$$M = \sum_{s \in Q_X} (\|b(\iota_{123}(s))\| + \|b(\iota_{012}(s))\|), \quad B = 2M + 2D,$$

$$K = 2(6|X| + 6)(B + 1).$$

Only the following members of these two finite embedded sets are used:

$$\begin{aligned} & x_{13}(1), x_{23}(1), x_{12}(t), x_{21}(t), \\ & x_{01}(1), x_{02}(1), x_{21}(t), x_{12}(t), \end{aligned} \quad t \in \{1\} \cup X.$$

Here the first row comes from the 123-embedding and the second from the 012-embedding. Every displayed element commutes with  $r = x_{03}(a)$  and normalizes  $N$ . Comparing the quasi-cocycle equations for  $sr$  and  $rs$ , then projecting to the moving subspace, gives for each displayed control element

$$\|\pi(s)w - w\| \leq \|\pi(s)b(r) - b(r)\| \leq 2\|b(s)\| + 2D \leq B.$$

Lemma 2.3 propagates these finitely many inequalities to the complete root planes in the two embeddings. In particular, for every  $c \in A$ ,

$$\|\pi(x_{13}(c))w - w\|, \quad \|\pi(x_{23}(c))w - w\|, \quad \|\pi(x_{01}(c))w - w\| \leq K.$$

The Steinberg relation

$$[I + cE_{01}, I + E_{13}] = I + cE_{03}$$

and the unitary commutator estimate give displacement at most  $4K$  under  $x_{03}(c)$ . Every element of  $N$  is a product of one element from each of  $X_{03}, X_{13}, X_{23}$ , so its displacement of  $w$  is at most  $6K$ . Since  $w \perp \mathcal{H}^N$ , this already bounds  $w$ . Indeed, the closed convex hull of the  $N$ -orbit of  $w$  has a unique vector of minimal norm; that vector is  $N$ -fixed. The convex hull stays inside  $(\mathcal{H}^N)^\perp$ , so its minimal vector is zero. On the other hand, the whole orbit, and hence its closed convex hull, lies in the closed ball of radius  $6K$  about  $w$ . Consequently  $\|w\| \leq 6K$ . Together with (10), this gives

$$\|b(I + aE_{03})\| \leq D + 12(6|X| + 6)(2M + 2D + 1),$$

uniformly in  $a \in A$ . □

**Remark 2.6** (A qualitative proof from Mimura's theorem). For qualitative boundedness, Mimura's relative theorem gives a shorter proof of Theorem 2.5. Split  $\mathcal{H} = \mathcal{H}^G \oplus (\mathcal{H}^G)^\perp$ . Mimura's relative theorem bounds the second projection on  $X_{03}$ . On the invariant summand the action is trivial, while  $r^2 = 1$  for  $r \in X_{03}$ ; the quasi-cocycle identities for 1 and  $r^2$  then give  $\|b_0(r)\| \leq D$ . The direct proof above complements this argument by recording an explicit bound and the quantitative rank-three propagation behind it.

### 3. COORDINATE-BLOCK FACTORIZATION

We now prove the main algebraic input. Write

$$e_{ij}(a) = I + aE_{ij} \quad (i \neq j)$$

for an elementary transvection.

**Lemma 3.1** (Five-move pivot). *Let  $R$  satisfy (1), let  $A \in \mathrm{GL}_n(R)$ , and fix distinct coordinates  $i, j$ . At most two elementary row operations and at most three elementary column operations produce a matrix  $B$  satisfying*

$$B_{jj} = 1, \quad B_{ij} = B_{ji} = 0.$$



*Proof.* We give the calculation, since it is the source of the uniform constant. Choose two distinct indices  $i, j$ . First arrange that the entry  $a = A_{ii}$  is nonzero. If it is zero, invertibility of  $A$  gives  $k$  with  $A_{ik} \neq 0$ ; right multiplication by  $e_{ki}(1)$  replaces  $A_{ii}$  by  $A_{ik}$ . This costs at most one preliminary column operation.

Write  $b = A_{ij}$ , and choose  $p, q \in R$  with  $paq = 1$ . Put

$$\rho = q(1 - pb), \quad F_1 = e_{ij}(\rho), \quad A_1 = AF_1, \quad b_1 = (A_1)_{ij}.$$

Then

$$b_1 = a\rho + b, \quad pb_1 = paq(1 - pb) + pb = 1.$$

Set

$$d_1 = (A_1)_{jj}, \quad w = (1 - d_1)p, \quad E_1 = e_{ji}(w), \quad A_2 = E_1A_1.$$

The  $(j, j)$ -entry is now

$$(A_2)_{jj} = d_1 + wb_1 = d_1 + (1 - d_1)pb_1 = 1.$$

Next set

$$E_2 = e_{ij}(-b_1), \quad A_3 = E_2A_2.$$

This clears  $(i, j)$  and preserves the new 1 at  $(j, j)$ . Finally, with  $c_3 = (A_3)_{ji}$ , put

$$F_2 = e_{ji}(-c_3), \quad A_4 = A_3F_2.$$

This clears  $(j, i)$ , again preserving  $(j, j) = 1$  and  $(i, j) = 0$ . Thus the four main moves use two row and two column transvections; including the possible preliminary column addition gives the asserted bounds 2 and 3. The calculation applies to arbitrary rings satisfying (1).  $\square$

*Proof of Theorem A.* Choose  $i \neq j$ . Lemma 3.1 gives elementary products  $E, F$ , represented by lists  $\ell, r$ , such that

$$B = EAF, \quad B_{jj} = 1, \quad B_{ij} = B_{ji} = 0,$$

with

$$|\ell| \leq 2, \quad |r| \leq 3.$$

For every  $k \neq j$ , right multiply by

$$e_{jk}(-B_{jk}).$$

These transvections have a common source column and distinct target columns, so each previously cleared entry stays zero. If  $T$  is their product and  $C = BT$ , then

$$C_{jc} = \begin{cases} 1, & c = j, \\ 0, & c \neq j. \end{cases}$$

The list  $t$  of right-clearing factors has length exactly  $n - 1$ .

For every  $k \neq j$ , left multiply by

$$e_{kj}(-C_{kj}).$$

Again the operations have distinct target rows and preserve the previously cleared entries. If  $S$  is their product and  $D = SC$ , then

$$D_{rc} = \begin{cases} 1, & r = c = j, \\ 0, & r = j, c \neq j, \\ 0, & r \neq j, c = j, \\ C_{rc}, & r, c \neq j. \end{cases}$$

Hence  $D \in \mathrm{GL}_{n-1}^{(j)}(R)$ . The list  $s$  of left-clearing factors also has length exactly  $n - 1$ .

Solving  $D = SEAF$  for  $A$  gives

$$A = E^{-1}S^{-1}DT^{-1}F^{-1}.$$

At the word level, inverse products are represented by reversing the factor lists and inverting every transvection. The inverse of  $e_{ab}(z)$  is  $e_{ab}(-z)$ , so all these factors remain elementary. The resulting word has the exact form

$$\ell_{\mathrm{rev}}^{-1} s_{\mathrm{rev}}^{-1} [D] t_{\mathrm{rev}}^{-1} r_{\mathrm{rev}}^{-1},$$

and therefore length at most

$$2 + (n - 1) + 1 + (n - 1) + 3 = 2n + 4.$$

□

**Remark 3.2.** The fixed coordinate  $j$  is chosen before  $A$ . Allowing the coordinate block to depend on  $A$  gives a different, weaker width statement; the fixed-coordinate form is what feeds the uniform quasi-cocycle globalization below.

#### 4. FROM RELATIVE CONTROL TO GLOBAL RIGIDITY

Let  $R$  be a finite-type  $\mathbb{F}_2$ -algebra. Choose a finite set  $X$  and a surjection

$$\mathbb{F}_2\langle X \rangle \twoheadrightarrow R.$$

Elementary matrices are functorial in the coefficient ring, so the root estimate of Theorem 2.5 descends to  $R$ . Coordinate permutations then give the same estimate for every root subgroup in rank four.

Two larger pieces are used in the assembly: a last-column subgroup and a last-row subgroup. The quasi-cocycle is bounded on both. The stabilized rank-three coordinate group normalizes these pieces. Kazhdan control for the rank-three group converts boundedness on the normalized root system into boundedness on the coordinate group, provided the ambient representation has no invariant vectors. This is the only place where the quotient by the trivial representation enters.

We record the relative-to-global step explicitly. For a subset  $S \subseteq G$ , say that quasi-cocycles are *bounded on  $S$*  if every quasi-cocycle into a unitary representation without invariant vectors has bounded restriction to  $S$ .

**Lemma 4.1** (Kazhdan globalization on a normalized set). *Let  $G$  have property  $(T)$ , let  $S \subseteq G$  generate  $G$ , and let  $H \leq G$ . Suppose*

$$h^{-1}Sh \subseteq S \quad (h \in H)$$

*and quasi-cocycles are bounded on  $S$ . Then  $(G, H)$  has relative property  $(TT)/T$ .*

*Proof.* Fix a unitary representation  $\pi$  of  $G$  without nonzero invariant vectors and a quasi-cocycle  $b$  of defect at most  $D$ . Choose a Kazhdan pair  $(K, \varepsilon)$ , and choose once and for all an  $S \cup S^{-1}$ -word for each  $k \in K$ . Let  $L$  be the sum of their lengths. If  $\|b(s)\| \leq C$  for  $s \in S$ , comparison of the quasi-cocycle identities for  $sh$  and  $h(h^{-1}sh)$  gives

$$\|\pi(s)b(h) - b(h)\| \leq 2C + 2D \quad (s \in S \cup S^{-1}, h \in H).$$

For an inverse letter, apply the comparison to its inverse and use  $\|\pi(s^{-1})v - v\| = \|\pi(s)v - v\|$ ; the constant remains  $2C + 2D$ . The displacement estimate for a product of unitaries therefore gives

$$\|\pi(k)b(h) - b(h)\| \leq L(2C + 2D) \quad (k \in K).$$

Since  $\pi$  has no invariant vectors, the Kazhdan inequality supplies some  $k \in K$  with  $\varepsilon\|b(h)\| \leq \|\pi(k)b(h) - b(h)\|$ . Thus  $\|b(h)\| \leq L(2C + 2D)/\varepsilon$ , uniformly in  $h \in H$ .  $\square$

**Lemma 4.2** (Leavitt rank equivalence). *If  $R$  contains a binary Leavitt family, then for all  $m, n \geq 2$  there is a group isomorphism*

$$E_m(R) \cong E_n(R).$$

*Proof.* Iterating the binary prefix code gives a ring isomorphism  $M_q(R) \cong R$  for every  $q \geq 2$ . Functoriality in the coefficient ring and block flattening give

$$E_m(R) \cong E_m(M_n(R)) \cong E_{mn}(R) \cong E_n(M_m(R)) \cong E_n(R).$$

The middle identifications are bidirectional on elementary groups. A block transvection decomposes into commuting scalar transvections after flattening. Conversely, a scalar transvection whose indices lie in different outer blocks is already a block transvection. If both indices lie in one outer block, choose a second outer block and use the Steinberg identity

$$e_{(i,k),(i,l)}(a) = [e_{(i,k),(j,k)}(a), e_{(j,k),(i,l)}(1)].$$

Thus block flattening identifies the full elementary subgroups in both directions, and the displayed chain restricts to the claimed group isomorphism.  $\square$

**Lemma 4.3** (Elementary equals general linear). *Assume that  $R$  has a binary Leavitt family, satisfies (1), and satisfies  $\text{diag}(u, 1) \in E_2(R)$  for every  $u \in R^\times$ . Then*

$$E_n(R) = \text{GL}_n(R) \quad (n \geq 2).$$

*Proof.* The single-sandwich elimination reduces an arbitrary element of  $\mathrm{GL}_2(R)$ , by elementary row and column operations, to  $\mathrm{diag}(u, 1)$  for some  $u \in R^\times$ . The diagonal hypothesis therefore gives  $\mathrm{GL}_2(R) = \mathrm{E}_2(R)$ .

The binary Leavitt family gives a coefficient isomorphism  $M_n(R) \cong R$  for every  $n \geq 2$ . Transporting the two ring-theoretic hypotheses across this isomorphism gives the same rank-two equality over  $M_n(R)$ . The bidirectional block flattening proved in Lemma 4.2 then gives

$$\mathrm{E}_2(M_n(R)) = \mathrm{GL}_2(M_n(R)) \implies \mathrm{E}_{2n}(R) = \mathrm{GL}_{2n}(R).$$

Equivalently, one may write the same flattening after interchanging the two finite index factors as

$$\mathrm{E}_n(M_2(R)) = \mathrm{GL}_n(M_2(R)).$$

Transport through  $M_2(R) \cong R$  now gives  $\mathrm{E}_n(R) = \mathrm{GL}_n(R)$ .  $\square$

**Proposition 4.4** (Rank-four assembly). *Under the hypotheses of Theorem B,  $\mathrm{E}_4(R)$  has property  $(TT)/T$ .*

*Proof.* Choose a finite free-algebra presentation of  $R$ . Theorem 2.5 descends through this quotient and gives relative  $(TT)/T$  for  $X_{03}(R)$ . Coordinate permutations give it for the six roots

$$X_{03}, X_{13}, X_{23}, X_{30}, X_{31}, X_{32}.$$

The assembly has the following dependency chain:

|                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\begin{aligned} &\text{six root bounds} \implies \text{last row and last column are bounded} \\ &\implies \text{the rank-three core is bounded} \implies \text{coordinate blocks are bounded} \\ &\xRightarrow{\text{Theorem A}} \text{the whole rank-four group is bounded.} \end{aligned}$ |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Here are the details. Put

$$C = X_{03}X_{13}X_{23}, \quad C^- = X_{30}X_{31}X_{32}.$$

The three roots in each product commute, so the quasi-cocycle identity makes  $b$  bounded on  $C$  and on  $C^-$ . The set  $S = C \cup C^-$  generates  $\mathrm{E}_4(R)$ : roots meeting the last coordinate lie in  $S$ , and every remaining root is a Steinberg commutator of one last-column and one last-row root.

Property  $(T)$  for  $\mathrm{E}_3(R)$  follows from Ershov–Jaikin–Zapirain. It can also be obtained by descending an explicit finite Kazhdan system for the free algebra. Lemma 4.2 transports property  $(T)$  to  $\mathrm{E}_4(R)$ . The stabilized rank-three subgroup  $H = \mathrm{E}_3(R)$  normalizes both  $C$  and  $C^-$ , hence normalizes  $S$ . Lemma 4.1 now proves that  $(\mathrm{E}_4(R), H)$  has relative property  $(TT)/T$ . Notice that the no-invariant-vectors hypothesis is used precisely in the Kazhdan inequality inside that lemma.

By Lemma 4.3, the elementary groups in ranks three and four are the full general linear groups. Under these identifications the stabilized core is exactly the coordinate block  $\mathrm{GL}_3^{(3)}(R)$ . Finally, coordinate permutations carry this relative statement to  $\mathrm{GL}_3^{(j)}(R)$  for every  $j \in \{0, 1, 2, 3\}$ .

Each elementary transvection lies in a coordinate block avoiding its two indices. Hence the union of one fixed coordinate block and all root subgroups is bounded for the quasi-cocycle. Theorem A writes every element of  $\mathrm{GL}_4(R)$  as a product of at most 12 elements of that union. Lemma 4.5 below, or directly the corresponding bounded-product lemma, makes the quasi-cocycle bounded on  $\mathrm{GL}_4(R)$ . Transporting through  $E_4(R) = \mathrm{GL}_4(R)$  proves the proposition.  $\square$

The passage from a bounded generating subset to the whole group uses the following elementary estimate.

**Lemma 4.5** (Products of controlled pieces). *Let  $b$  be a quasi-cocycle of defect at most  $D$ , and suppose  $\|b(g_i)\| \leq C$  for a list  $(g_1, \dots, g_m)$ . Then*

$$\|b(g_1 \cdots g_m)\| \leq mC + (m+1)D.$$

*Proof.* Induct on the list. The empty-list case is  $\|b(1)\| \leq D$ . For a nonempty list, apply (2) to the head and the product of the tail, use that the representation is isometric, and invoke the inductive bound. The displayed estimate is a slightly loose uniform form that also covers the empty list.  $\square$

*Proof of Theorem B.* Proposition 4.4 gives the result in rank four. For arbitrary  $n \geq 2$ , Lemma 4.2 supplies an explicit group equivalence  $E_n(R) \cong E_4(R)$ . Property  $(TT)/T$  is invariant under group equivalence, so the rank-four conclusion transports to rank  $n$ .  $\square$

## 5. THE BINARY LEAVITT SPECIALIZATION

Let  $L$  be the binary Leavitt algebra [5],

$$L = L_{\mathbb{F}_2}(1, 2) = \mathbb{F}_2\langle x_0, x_1, y_0, y_1 \rangle / (y_i x_j - \delta_{ij}, x_0 y_0 + x_1 y_1 - 1).$$

The defining generators themselves form a binary Leavitt family, and the presentation makes  $L$  a finite-type  $\mathbb{F}_2$ -algebra. The graph criterion of Abrams–Aranda Pino shows that  $L$  is purely infinite simple [1, Theorem 11]; Ara–Goodearl–Pardo’s Theorem 1.6 then gives the single-sandwich property. It remains to discuss the unstable diagonal condition.

**Proposition 5.1** (Unstable diagonal reduction). *For every  $u \in L^\times$ ,*

$$\mathrm{diag}(u, 1) \in E_2(L).$$

*Consequently*  $\mathrm{GL}_2(L) = E_2(L)$ .

*Proof.* Ara–Brustenga–Cortiñas give, for a regular coefficient ring, the exact sequence determined by  $1 - N_E^t$  [2]. For the graph with one vertex and two loops this map is multiplication by  $-1$ , hence

$$K_i(L) = 0$$

for every  $i \in \mathbb{Z}$ , in particular in degree one.

The GE criterion of Menal–Moncasi applies to unital purely infinite simple rings, as recalled by Ara–Goodearl–Pardo in the proof of their Theorem 2.4 [10, 3]. Ara–Goodearl–Pardo’s theorem also identifies

$$K_1(R) \cong R_{\text{ab}}^\times$$

for such rings [3, Theorem 2.4]. Thus  $K_1(L) = 0$  makes each unit a product of commutators. The standard  $2 \times 2$  Whitehead identity places  $\text{diag}([a, b], 1)$  in  $E_2(L)$ ; the GE property then gives  $\text{GL}_2(L) = E_2(L)$ , proving Proposition 5.1.  $\square$

*Proof of Theorem C.* The binary Leavitt algebra is a nontrivial finite-type  $\mathbb{F}_2$ -algebra, its canonical generators form a binary Leavitt family, pure infiniteness supplies (1), and Proposition 5.1 supplies the elementary diagonal condition. Theorem B applies.  $\square$

## 6. FURTHER DIRECTIONS

The characteristic-two hypothesis enters most visibly in (10), where every root element has order two. Over a field of characteristic  $p$ , the corresponding root has order  $p$ , and iterating the quasi-cocycle identity gives

$$b(1) \approx b(r) + \pi(r)b(r) + \cdots + \pi(r)^{p-1}b(r).$$

On the root-fixed space this is  $p$  times the fixed projection of  $b(r)$ , with error at most  $(p-1)D$ . Thus the fixed-space argument has a plausible finite-characteristic analogue. Extending the theorem requires uniform replacements for the characteristic-two rank-three control and the sign-free coordinate symmetries.

The algebraic factorization theorem suggests a separate question. For which natural classes of simple or purely infinite rings does the single-sandwich condition hold, and when can the coordinate block in Theorem A itself be controlled by elementary subgroups? A satisfactory answer would extend the present synthesis from the Leavitt setting to a broader rigidity criterion.

## APPENDIX A. FORMAL THEOREM CORRESPONDENCE

The accompanying Lean formalization is available in the [project repository](#). The table below records the declaration corresponding to each principal statement. Except where indicated, declarations are in the `PropertyTTPaper` namespace.

| Paper statement                                                | Lean declaration                                                                                                                                                                                              |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem <a href="#">A</a>                                      | <code>coordinateBlock_factorization</code>                                                                                                                                                                    |
| Lemma <a href="#">3.1</a>                                      | <code>fiveMove_pivot</code>                                                                                                                                                                                   |
| Lemma <a href="#">2.2</a>                                      | <code>freeCharacteristicTwo_planeEnergyEstimate</code>                                                                                                                                                        |
| Lemma <a href="#">2.3</a>                                      | <code>freeCharacteristicTwo_localizedPlaneEstimate</code>                                                                                                                                                     |
| Lemma <a href="#">2.4</a>                                      | <code>freeCharacteristicTwo_inverseTranspose_root,</code><br><code>freeCharacteristicTwo_rowEmbedding_</code><br><code>controlRoot, freeCharacteristicTwo_</code><br><code>rowEmbedding_elementaryRoot</code> |
| Theorem <a href="#">2.5</a>                                    | <code>freeCharacteristicTwo_root_hasRelativeTT</code>                                                                                                                                                         |
| Lemma <a href="#">4.1</a>                                      | <code>kazhdan_normalizedSet_globalization</code>                                                                                                                                                              |
| Lemmas <a href="#">4.2</a> and <a href="#">4.3</a>             | <code>leavitt_elementaryRankEquivalence,</code><br><code>elementaryBlockFlattening,</code><br><code>elementaryGroup_eq_generalLinear</code>                                                                   |
| Proposition <a href="#">4.4</a>                                | <code>finiteFreeQuotient_rankFour_hasTTmodT</code>                                                                                                                                                            |
| Lemma <a href="#">4.5</a>                                      | <code>quasiCocycle_list_product_bound</code>                                                                                                                                                                  |
| Theorem <a href="#">B</a>                                      | <code>finiteType_elementaryGroup_hasTTmodT</code>                                                                                                                                                             |
| Proposition <a href="#">5.1</a>                                | <code>KOnePaper.diagUnit_mem_elementary,</code><br><code>KOnePaper.elementaryGroup_two_eq_top</code>                                                                                                          |
| Theorem <a href="#">C</a> and<br>Corollary <a href="#">1.1</a> | <code>binaryLeavitt_elementaryGroup_hasTTmodT,</code><br><code>binaryLeavitt_elementaryGroup_hasTTmodT_and_</code><br><code>not_isSofic</code>                                                                |

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